

MSc Project Proposal



MSc Data Science and AI Project Proposal

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Programme of Study

Data Science

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The All-Rounder

An End-to-End Data Engineering and Predictive Framework for Demand Forecasting and Inventory Optimization in Pharmaceutical Distribution

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1. Abstract

The research goal of this project is to optimize supply chain efficiencies within the wholesale pharmaceutical distribution sector through the development of an automated, end-to-end data engineering pipeline and predictive AI framework. Utilizing a robust 5-year historical dataset containing transactional order ledgers, this study builds scalable architectures to systematically clean, process, and ingest high-dimensional stock data. Advanced machine learning models—specifically gradient boosting algorithms (e.g., XGBoost) and classical time-series models—will be implemented to accurately forecast SKU-level demand variations. The outputs will feed directly into an inventory optimization layer to dynamically compute safety stock and automated reorder points (*ROP*). The ultimate objective is to mitigate the dual logistical constraints of stockouts and product expiration waste, providing an actionable operational roadmap for modern pharmaceutical logistics.

2. Introduction

Modern pharmaceutical supply chains are tightly regulated networks that face distinct logistical challenges, such as volatile demand fluctuations, strict temperature controls, and finite product shelf lives. Inefficient inventory management leads either to catastrophic stockouts of life-saving medicines or expensive capital traps caused by expired overstocks. Traditional forecasting approaches often rely on basic rolling averages or static heuristic methods that fail to adapt to complex seasonal disease spikes, geographic anomalies, and dynamic order patterns.

This study introduces a comprehensive data-driven solution that shifts wholesale pharmacy distribution from reactive stocking to proactive, AI-guided replenishment. By utilizing an extensive 5-year ledger of historical order data, this project addresses structural inefficiencies in matching stock levels to regional health demands through advanced data pipelines and optimized time-series predictions.

- **Aims and Objectives**

- Aim 1: Build a scalable, automated data engineering pipeline for processing large-scale transactional pharmaceutical data.**

- 1.1:** Clean, normalize, and schema-map 5 years of historical order ledgers to resolve missing entries, clerical errors, and inconsistencies.
 - 1.2:** Aggregate data structurally by timestamp, geographic location, client tier, and product category (aligning with therapeutic classifications).
 - 1.3:** Build an optimized ETL/ELT architecture to automate the transition of raw data into analytical features.
 - 1.4:** Conduct Exploratory Data Analysis (EDA) to map non-linear demand trends, outliers, and seasonal variances across product tiers.

Aim 2: Develop and validate predictive machine learning frameworks for multi-level demand forecasting and inventory replenishment.

- 2.1:** Implement and evaluate classical time-series baselines (such as SARIMA) against advanced machine learning algorithms (such as XGBoost/CatBoost).
- 2.2:** Engineer predictive features incorporating temporal factors (e.g., month, season indicators) and historical lags.
- 2.3:** Translate forecast parameters into mathematical safety stock metrics and automated dynamic Reorder Points (*ROP*).
- 2.4:** Build an evaluation interface/dashboard to visually validate model outputs against actual historical hold patterns.

- **Contribution of the Project**

This project contributes academically by providing an empirical blueprint on how end-to-end data engineering pipelines directly operationalize machine learning models inside the logistics sector—bridging the gap between isolated algorithmic design and real-world execution. Mechanically, the framework helps wholesale distributors lower holding expenditures, prevent waste due to expired drug stocks, and secure consistent, uninterrupted medicine access for downstream clinics and pharmacies.

3. State of the Art Review

The academic landscape surrounding supply chain management increasingly characterizes the pharmaceutical sector as highly volatile due to epidemiological waves and complex regulatory governance. Historically, statistical time-series forecasting frameworks like Autoregressive Integrated Moving Average (ARIMA) and its seasonal variant (SARIMA) formed the baseline for inventory planning. While computationally lean and highly interpretable, literature demonstrates that these linear models perform poorly when dealing with intermittent, sparse order patterns typical of specialized or high-cost medications.

To counter these limitations, recent machine learning literature highlights gradient-boosted decision trees (such as XGBoost) and deep learning long short-term memory (LSTM) networks as superior frameworks for capturing sharp, non-linear demand shifts. Supervised machine learning allows for the integration of multidimensional exogenous variables—such as regional demographics and historical supplier delays—significantly reducing Symmetric Mean Absolute Percentage Error (SMAPE) relative to linear models.

However, a distinct gap persists within current research: the majority of academic papers analyze model tuning inside highly sanitized environments using isolated notebook files. There is a lack of focus on the **end-to-end data engineering workflows** necessary to handle structural data drift, stream ingestion, and the conversion of predictive arrays into actionable inventory logic (such as safety stock variations). This thesis directly bridges

this gap by demonstrating an operationalized system using a continuous 5-year historical ledger.

4. Methods

This study adopts an empirical, data-driven methodology categorized into three major components:

- **Data Acquisition & Engineering:** The primary data source consists of 5 years of historical order records from a wholesale pharmaceutical distributor. The data will undergo systematic preprocessing (imputation of null entries, duplicate elimination, and formatting standardization). Features will be engineered by creating time-lag indicators (e.g., previous 7-day, 30-day volumes) and categorical flags matching core drug families.
- **Algorithmic Modeling Framework:** The forecasting task will be modeled as a supervised regression problem. We will deploy and tune an **XGBoost Regressor** pipeline alongside a **SARIMA** baseline. The optimization objective will prioritize reducing error parameters across varying operational horizons (weekly vs. monthly predictions).
- **Inventory Optimization Logic:** The predicted demand values (D) and calculated lead times (L) will be programmatically fed into safety stock equations to calculate dynamic Reorder Points (ROP) via:

$$ROP = (Average\ Daily\ Demand \times Lead\ Time) + Safety\ Stock$$

This ensures the predictive model translates cleanly into real-world business actions.

5. Proposed Work

5.1 Intended contribution of the work

The primary deliverable is an automated pipeline that accepts historical, multi-product transactional inputs and transforms them into predictive stocking triggers. It integrates content-based grouping, where drug families with low individual frequencies are aggregated into broader categories to overcome data sparsity challenges during model training.

5.2 Benefits of the proposed work

The primary beneficiaries are wholesale pharmaceutical managers, logistical planners, and healthcare institutions. Implementing this system minimizes capital lock-up in sluggish inventories, decreases disposal overheads from expired pharmaceuticals, and reduces the probability of regional drug shortages.

5.3 Procedures and activities of the proposed work

Objective	Activity	Milestone
Data Architecture	Ingest, clean, and map the 5-year historical distribution dataset.	Preprocessed dataset and feature schema finalized.
Model Prototyping	Train, optimize, and cross-validate ARIMA and XGBoost architectures.	Completed model training runs with optimized hyper-parameters.
Inventory Synthesis	Code the dynamic reorder layer using model prediction coefficients.	Inventory engine logic fully validated.
Operational Validation	Evaluate error distributions and display metrics via an analytical layout.	Analytical visualization completed and documented.

5.4 Evaluation of the project

- **Quantitative Model Evaluation:** Predictive accuracy will be rigorously evaluated using root mean squared error ($RMSE$), mean absolute error (MAE), and R^2 scores across training sets.
- **Operational Simulation:** The inventory optimization logic will be back-tested against the historical ledger to measure simulated reductions in overstock volumes and stockout events.
- **Process Review:** Regular development check-ins alongside academic advisors will verify architectural adherence to scalable, best-practice data engineering design principles.

5.5 Resources needed

- **Software Environment:** Python 3.x ecosystem (Pandas, NumPy, Scikit-Learn, XGBoost, Stats models).
- **Database Infrastructure:** Local or cloud-based relational database instances (PostgreSQL / SQLite).
- **Hardware Compute:** Standard workstation with multi-core CPU capabilities for parallelizing hyper-parameter grid searches.

5.6 Access to participants and clients

The project relies on an existing, anonymized 5-year transaction history log, bypassing the need for live human participant recruitment.

5.7 Ethical Aspects

The transaction ledger is strictly restricted to business-to-business (B2B) operational metrics (SKU counts, product identifiers, anonymized client location codes, and transaction timestamps). It contains **no identifiable patient information, personal medical prescriptions, or sensitive individual records**, ensuring complete alignment with standard university data safety governance.

6 Work plan and schedule

The plan is created on **Smartsheet** a cloud project management tool, and an **excel export** of the **WBS** and **Gantt chart** are provided to illustrate the timeline and milestones.

Phase/Objective	Activities	Milestone
Phase1: Initiation		
	Project scope definition and architecture mapping	Proposal
	Initial database structural exploration	
Phase2: Literature Review		
	Research time-series and demand forecasting papers	Proposal and thesis
	Synthesize algorithmic comparison summaries	
Phase3: Requirements Analysis & Design		
	Dataset collection phase	Feature Warehouse Built
	Data pipeline schema mapping and ETL design	
	Data cleansing and missing vector management	
Phase4: Development		
	Feature engineering & lag creation	Fully Trained Forecasting Engine
	Baseline (SARIMA) vs ML (XGBoost) modeling	
Phase5: Documentation & Graded Deliverables		
	proposal draft	Thesis
	final proposal	
	formal project proposal	
	Ch1 introduction partial and ch2 literature review	

	submission, approval, and feedback	
	final report	
	viva voice plus poster	

7 Conclusions

This proposal outlines a robust, dual-discipline project that directly satisfies the advanced competencies of an MSc in Data Engineering and AI. By converting 5 years of historical pharmaceutical distribution records into a live, predictive optimization mechanism, the research addresses real-world industrial inefficiencies. The systematic structure detailed within this document ensures clear execution and straightforward tracking as the project advances toward its final delivery.

8 Bibliography

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9 Appendixes

- WBS

Health	Primary	Start Date	End Date	% Completion
Project Information				
Yellow	Project Plan	25-Jun-26	01-Jul-26	60%
Green	Phase 1: Project Initiation & Data Exploration	05-Jul-26	20-Jul-26	60%
Green	Discovery on pharmaceutical supply chain solutions & research	06-Jul-26	17-Jul-26	100%
Green	Explore data sources and database constraints (5-year invoice ledger)	08-Jul-26	24-Jul-26	80%
Green	Scope definition and final parameter mapping	09-Jul-26	25-Jul-26	50%
Green	Phase:2 Literature review	20-Jun-26	09-Jul-26	100%
Green	Research time-series & GBDT papers on Google Scholar / IEEE	21-Jun-26	09-Jul-26	100%
Green	Research supply chain operationalization architectures	22-Jun-26	09-Jul-26	100%
Blue	Phase:3 Requirements analysis & Design	25-Jul-26	20-Aug-26	0%
Blue	Database initialization and ingestion schema configuration (SQL)	25-Jul-26	05-Aug-26	0%
Blue	Prototype ETL logic—prioritize SKU-level description classifications	30-Jul-26	10-Aug-26	0%
Blue	Prototype temporal flag mapping (Date vectors and QTY scaling parameters)	06-Aug-26	16-Aug-26	0%
Blue	Prototype baseline exploratory tracking charts	15-Aug-26	20-Aug-26	0%
Blue	Phase:4 Design	25-Aug-26	05-Sep-26	0%

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Blue	Design system context and automated data engineering pipeline diagrams	25-Aug-26	30-Aug-26	0%
Blue	Model execution pipeline blueprint architecture	31-Aug-26	05-Sep-26	0%
Blue	Phase:5 Development	06-Sep-26	27-Sep-26	0%
Blue	Complete pipeline automation script assembly	06-Sep-26	11-Sep-26	0%
Blue	Train and evaluate baseline time-series frameworks (SARIMA)	12-Sep-26	18-Sep-26	0%
Blue	Program, evaluate, and cross-validate the supervised XGBoost Regressor	19-Sep-26	25-Sep-26	0%
Blue	Interface creation (Power BI dashboard layer to visualize forecast variances)	25-Sep-26	27-Sep-26	0%
Green	Phase:6 Documentation & Graded Deliverables	25-Jun-26	01-Oct-26	10%
Green	Initial proposal draft compilation	25-Jun-26	01-Jul-26	100%
Green	Final project proposal generation and layout assembly	01-Jul-26	10-Jul-26	100%
Green	Chapter 1 (Introduction) and Chapter 2 (Literature Review) submission	20-Jun-26	09-Jul-26	100%
Blue	Academic mid-point review evaluation and milestone review check	10-Aug-26	10-Aug-26	0%
Blue	Final comprehensive report compilation and technical write-up	28-Sep-26	28-Sep-26	0%
Blue	Viva voce presentation practice and final poster delivery	01-Oct-26	01-Oct-26	0%



Literature Review Document

Course Title: MSc Data Science and AI

Project Title: An End-to-End Data Engineering and Predictive Framework for Demand Forecasting and Inventory Optimization in Pharmaceutical Distribution

Student Name: Shiyas Rasi

Student ID: M01087756

1. Introduction

Pharmaceutical distribution operates under conditions that make demand forecasting and inventory control unusually difficult. Demand can be seasonal, intermittent, geographically uneven, and sensitive to epidemiological events, while inventory decisions are constrained by product shelf life, service-level expectations, lead-time uncertainty, and the financial cost of holding excess stock. In this context, inaccurate forecasts can contribute to both stockouts and avoidable expiry-related waste. The problem is therefore not only to improve predictive accuracy, but also to translate forecasts into reliable replenishment decisions.

The forecasting literature has evolved from classical statistical methods, including exponential smoothing and ARIMA-family models, towards machine-learning and deep-learning approaches that can represent nonlinear relationships and use a wider set of explanatory variables. Research on pharmaceutical demand has compared multiple statistical and machine-learning approaches, while the broader intermittent-demand literature shows that sparse and irregular demand requires careful model selection, evaluation, and inventory-aware validation (Nikolopoulos et al., 2016; Türkmen et al., 2021; Li et al., 2022; Yani et al., 2023).

However, predictive performance alone does not establish operational value. A forecasting model depends on the quality, consistency, and timeliness of upstream data, while its outputs must be integrated with downstream inventory rules such as safety stock and reorder points. This review therefore examines four connected themes: pharmaceutical demand characteristics; classical forecasting methods; machine-learning and deep-learning approaches; and the operational integration of forecasting, data engineering, and inventory optimization. The purpose is to establish the conceptual and methodological basis for the proposed Pharma-Flow framework and to identify a defensible research gap.

2. Literature Review Methodology

A structured literature review approach was used to identify research relevant to pharmaceutical demand forecasting, intermittent demand, machine learning, inventory optimization, and data-engineering operationalization. Searches were designed around combinations of terms such as “pharmaceutical demand forecasting”, “drug demand prediction”, “intermittent demand forecasting”, “ARIMA”, “SARIMA”, “XGBoost”, “LSTM”, “inventory optimization”, “safety stock”, “reorder point”, “ETL pipeline”, and “MLOps”. Sources were considered from academic databases and indexing services including Google Scholar, IEEE Xplore, ScienceDirect, and publisher databases.

The inclusion criteria prioritized peer-reviewed journal articles and conference papers that: (1) addressed demand forecasting, inventory planning, or supply-chain prediction; (2) reported a quantitative modelling method; (3) provided a relevant comparison of statistical, machine-learning, or deep-learning techniques; or (4) examined intermittent demand or the connection between forecasting and inventory decisions. Technical documentation was considered only where it contributed directly to architectural or implementation considerations. Purely conceptual supply-chain discussions without a forecasting, data, or inventory-control component were excluded.

For each selected study, the review considered the application domain, data characteristics, forecasting horizon, model family, feature design, evaluation metrics, and relationship between forecast outputs and inventory decisions. The synthesis is thematic rather than a simple paper-by-paper summary. This is important because the proposed project combines several disciplines—data

engineering, forecasting, and inventory optimization—and the research gap lies partly in the integration between these areas.

3. Evolution of Demand Forecasting Research

3.1 Classical statistical forecasting

Classical forecasting methods remain important because they provide transparent and computationally efficient baselines. Naïve forecasts, moving averages, exponential smoothing, ARIMA, and seasonal ARIMA models are widely used where historical demand exhibits stable level, trend, and seasonality. In pharmaceutical forecasting, Nikolopoulos et al. (2016) compared a range of methods for branded and generic pharmaceutical series, including ARIMA, exponential smoothing, naïve methods, regression, and diffusion models. This work illustrates an important point: no single forecasting family is universally optimal, and pharmaceutical series can differ substantially according to product lifecycle and market behaviour.

The principal strength of ARIMA-family models is their explicit treatment of autocorrelation and, in seasonal variants, recurring temporal structure. They are interpretable and suitable as benchmarks. Their limitations become more visible when demand is highly sparse, lumpy, or influenced by nonlinear interactions among multiple variables. In such cases, aggregation level and forecast horizon can materially change performance. Consequently, the proposed project should retain SARIMA as a baseline rather than assume in advance that a machine-learning model will always be superior.

3.2 Intermittent and lumpy demand

Intermittent demand is especially relevant to pharmaceutical distribution because some SKUs may have many zero-demand periods followed by irregular positive orders. Standard accuracy measures and conventional forecasting models can behave poorly in this setting. Türkmen et al. (2021) proposed a renewal-process perspective that generalizes Croston-type approaches and supports more flexible modelling of demand arrivals. Rožanec and Mladenović (2021) similarly reframed lumpy-demand forecasting as two related tasks: predicting whether demand will occur and estimating demand size.

Li et al. (2022) examined feature-based forecast combinations for intermittent demand and emphasized not only point accuracy but also bias, quantile forecasting, computational efficiency, and inventory implications. These findings are relevant to PharmaFlow because a single global regression model trained across all SKUs may conceal important differences between fast-moving and intermittent products. A stronger design is therefore to classify demand patterns or product segments first, then compare forecasting performance by segment and evaluate whether grouping low-frequency products improves or weakens operational results.

3.3 Machine learning and gradient boosting

Machine-learning methods can incorporate calendar effects, lagged demand, rolling statistics, product attributes, customer or region information, and other exogenous variables. Tree-based ensemble methods are particularly attractive for tabular transactional data because they can model nonlinear relationships and interactions without requiring the same distributional assumptions as classical time-series models. XGBoost is a regularized gradient-boosting framework designed for efficient and scalable tree learning (Chen and Guestrin, 2016).

In pharmaceutical demand research, recent comparative studies increasingly evaluate machine-learning approaches alongside statistical baselines. Yani et al. (2023) examined demand-forecasting accuracy in the pharmaceutical supply chain and highlighted the relevance of machine-learning algorithms to this domain. Ascensão (2023) compared XGBoost, LSTM, and Prophet using pharmaceutical sales data and a structured CRISP-DM process. Such studies support the inclusion of XGBoost in the proposed framework, but they do not justify treating it as automatically superior. Performance depends on data granularity, feature engineering, horizon, product heterogeneity, and validation design.

For PharmaFlow, XGBoost is methodologically suitable because the source data are transactional and can be transformed into lagged and rolling features. Nevertheless, leakage prevention is essential. Random train-test splitting would be inappropriate for time-dependent demand data because future observations could indirectly influence model training. The project should use chronological holdout testing or rolling-origin evaluation, with all lag and rolling-window features computed using only information available before each forecast origin.

3.4 Deep learning approaches

LSTM networks and related deep-learning architectures are designed to learn temporal dependencies and have been applied to demand forecasting. Their potential advantage is the ability to represent complex sequential relationships, particularly when large and sufficiently regular datasets are available. However, deep-learning models generally introduce greater computational cost, tuning complexity, and lower interpretability than many classical or tree-based alternatives.

The literature does not support a universal claim that LSTM is either always better or always worse than gradient boosting. Comparative performance is context-dependent. For a medium-scale MSc project based on five years of B2B transaction data, the most defensible approach is to use a strong statistical baseline and one or more well-justified machine-learning models, while treating deep learning as an optional extension if data volume and time permit. This keeps the evaluation aligned with the research objectives rather than expanding the model set without clear analytical value.

4. From Forecast Accuracy to Inventory Decisions

A major theme emerging from the literature is that forecasting and inventory control should not be evaluated independently. A model with a lower average forecast error does not necessarily produce the lowest stockout rate or holding cost. Forecast bias, lead-time uncertainty, service-level targets, and demand variability all influence inventory performance. Research on intermittent demand also demonstrates the value of examining inventory implications rather than relying exclusively on aggregate error measures (Li et al., 2022).

The proposed project therefore has a strong opportunity to evaluate two layers of performance. The first is predictive performance, using metrics such as MAE, RMSE, and a percentage-based measure selected carefully for zero-demand series. The second is operational performance, using historical back-testing to compare stockout frequency, service level or fill rate, average inventory, excess stock, and expiry-risk proxies where expiry information is available. This dual evaluation would make the contribution stronger than a study that reports forecasting error alone.

The inventory layer should also define its assumptions explicitly. A reorder point can be represented conceptually as expected demand during lead time plus safety stock. The exact safety-stock formulation depends on whether uncertainty is attributed to demand, lead time, or both.

PharmaFlow should therefore document the chosen service level, lead-time assumptions, demand variability window, and replenishment review policy. Without these assumptions, the connection between prediction and replenishment would remain under-specified.

5. Data Engineering and Operationalization

Forecasting research often begins with a modelling-ready dataset, whereas operational systems must create that dataset repeatedly and consistently. In a real distribution ledger, likely engineering challenges include duplicate transactions, inconsistent product identifiers, missing timestamps, returns or cancellations, unit-of-measure inconsistencies, changes in product master data, and irregular transaction timing. These issues directly affect feature quality and model validity.

The proposed framework addresses this problem by designing a reproducible pipeline from raw transactional data to analytical features and forecasts. A relational database such as PostgreSQL can support staging, cleaned, and analytical layers. The pipeline should include data-quality checks, documented transformation rules, reproducible feature generation, model-version metadata, and a clear separation between training and inference datasets. These components are more academically meaningful when they are evaluated rather than merely implemented.

Accordingly, the data-engineering contribution should be assessed using measurable criteria such as pipeline reproducibility, data-quality rule coverage, processing time, failure handling, and the ability to rerun the workflow on new data without manual restructuring. This framing strengthens the claimed operationalization gap: the contribution is not simply the use of PostgreSQL or ETL, but the empirical integration and evaluation of a repeatable data-to-decision workflow.

6. Critical Synthesis and Research Gap

The reviewed literature provides substantial evidence for three separate propositions. First, pharmaceutical and related supply-chain demand can exhibit heterogeneous patterns that require more than one forecasting strategy. Second, machine-learning methods can exploit nonlinear relationships and engineered features, but their advantage over statistical methods is data- and context-dependent. Third, intermittent-demand research shows that model evaluation should consider demand occurrence, bias, forecast uncertainty, and inventory consequences.

The gap addressed by this project should therefore be stated carefully. It would be too broad to claim that existing research systematically ignores data pipelines or inventory integration. A more defensible gap is that comparatively fewer empirical studies evaluate, within one reproducible pharmaceutical-distribution workflow, the full chain from raw multi-SKU transactional data, through governed preprocessing and feature engineering, to time-aware model comparison and inventory-policy back-testing.

PharmaFlow addresses this integration gap through four linked contributions: a reproducible ETL/ELT pipeline for five years of multi-SKU B2B transaction data; a time-aware comparison of SARIMA and gradient-boosting models; segment-aware analysis of heterogeneous and intermittent demand; and an inventory simulation layer that converts forecasts into replenishment decisions and evaluates operational outcomes. The novelty is therefore located in integration, reproducibility, and operational evaluation rather than in proposing a new forecasting algorithm.

7. Implications for the Proposed Methodology

The literature review leads to several concrete methodological decisions. The project should first profile demand at SKU and aggregate levels, including zero-demand frequency, intermittency, seasonality, trend, and volume concentration. Product segmentation should be considered before model training because high-volume and intermittent items may require different treatment.

Second, model evaluation should use time-series cross-validation or rolling-origin evaluation. SARIMA should serve as an interpretable baseline, while XGBoost or CatBoost can be evaluated using lagged demand, rolling statistics, calendar variables, and relevant product or client features. Hyperparameter tuning must be nested within the temporal validation design to avoid optimistic estimates.

Third, evaluation metrics should be chosen with the data characteristics in mind. MAE and RMSE are useful scale-dependent measures, but MAPE can be unstable or undefined when actual demand is zero. Where intermittent demand is substantial, the study should consider scaled or zero-safe measures and report results by demand segment rather than only as a global average.

Finally, inventory back-testing should compare a baseline replenishment policy with forecast-driven policies under documented lead-time and service-level assumptions. The final dashboard should communicate both forecast quality and operational outcomes, thereby supporting decision-making rather than functioning only as a visualization of predicted values.

8. Conclusion

The literature indicates a clear progression from classical statistical forecasting towards machine-learning and deep-learning approaches capable of representing nonlinear demand patterns and incorporating richer features. Nevertheless, the evidence also shows that model superiority is conditional: data characteristics, intermittency, forecast horizon, feature design, and validation strategy all affect results. For pharmaceutical distribution, forecasting should therefore be treated as one component of a broader inventory decision system.

The proposed Pharma-Flow project is well positioned to contribute through an integrated and reproducible framework that links raw transactional data, data engineering, demand segmentation, time-aware model comparison, and inventory-policy evaluation. The strongest research contribution is not a claim that one algorithm universally outperforms another, but an empirical demonstration of how forecasting methods can be operationalized and assessed end to end on real multi-SKU pharmaceutical distribution data. This revised framing aligns the literature review more closely with the project proposal and provides a clearer basis for the methods, evaluation plan, and intended contribution.

9. References

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